# Starfish hoodie

size 128 · wide fit, Sweat / French Terry (25 % stretch)

5 parts · 16 steps · medium · about 2.5 h



Front view



Back view

**Sewn with: Sewing machine**



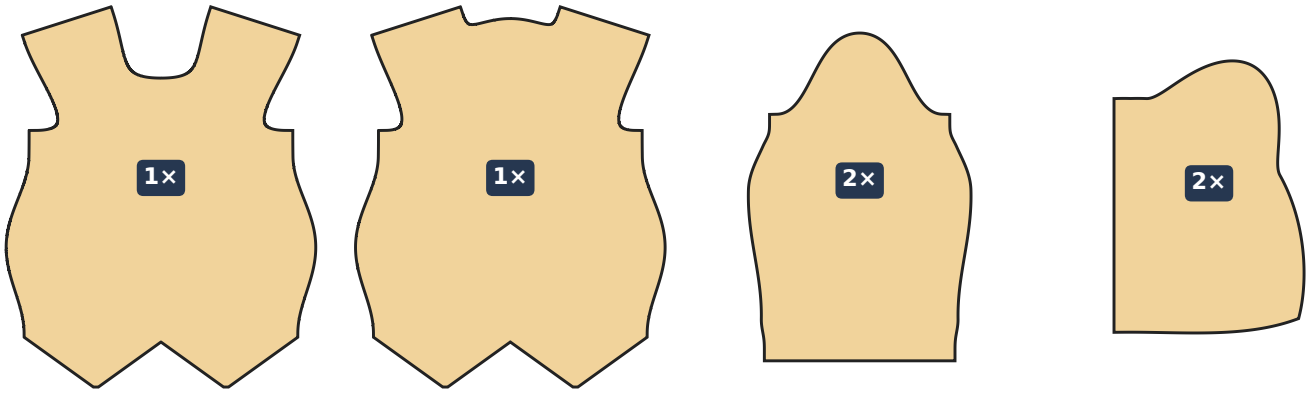
Sewing machine



Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

# Parts




A · Front (cut on fold)

B · Back (cut on fold)

C · Sleeve

E · Hood




D · Sleeve cuff

Seam allowance (included): 1.0 cm · turned-up hems 2.5 cm

# Material

**Main fabric**
1.05 m at 150 cm width

**Rib knit**
0.15 m at 70 cm width

**Sewing machine**
narrow zigzag 1 × 2.5 mm (or overlocker); twin needle 4 mm, stitch 3 mm; wide zigzag 4 × 1.5 mm over the edge

**Iron**
steam

**Polyester sewing thread**
In the colour of the fabric.

**Clear elastic or twill tape**
Sewn into the shoulder seams, so they do not stretch out.

**Pins or clips**
Pin across the seam line.

**Chalk or a fabric pen**
For notches, labels and placement lines.

## Colours in this booklet

**Main fabric**
right side · wrong side

**Rib knit**
right side · wrong side

Sew now (machine)

Sew by hand

Already sewn

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left sleeve is on the right when you look at the child.

# Basics: Sewing machine

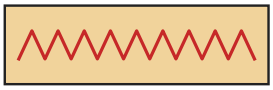
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

## Basics: the sewing machine



### Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



### Zigzag

Knit seams (narrow) and finishing the edges (wide).

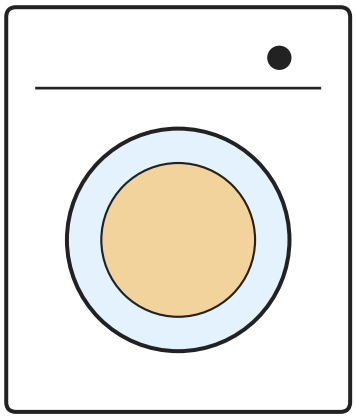


### Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

## 1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



30–40°

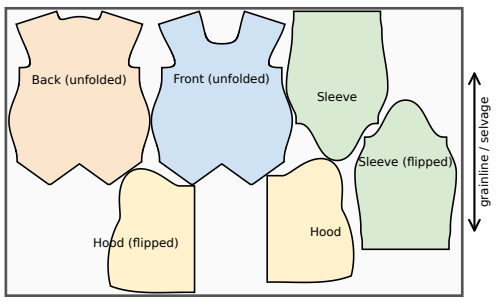
## 2 Cut out all pieces

Scissors

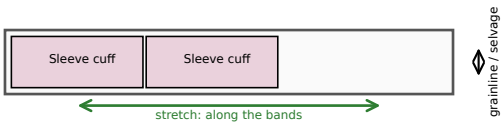
1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.
3. Rib knit: the ribs run along the grainline, so every band stretches along its length (across the fabric).

Tip: Single layer: cut the second piece of a pair with the paper piece turned over (printed side down).

Main fabric · 150 cm · single layer




Rib knit · 70 cm · single layer

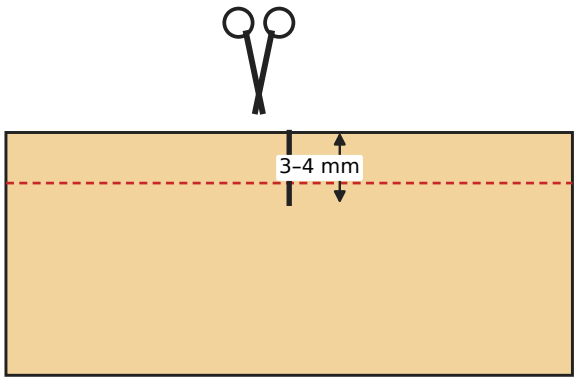


## 3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

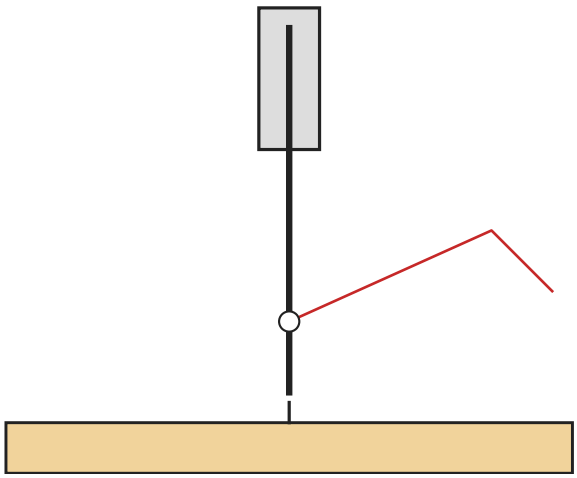
⚠ Important: Never deeper than half the seam allowance.




## 4 Set up the machine

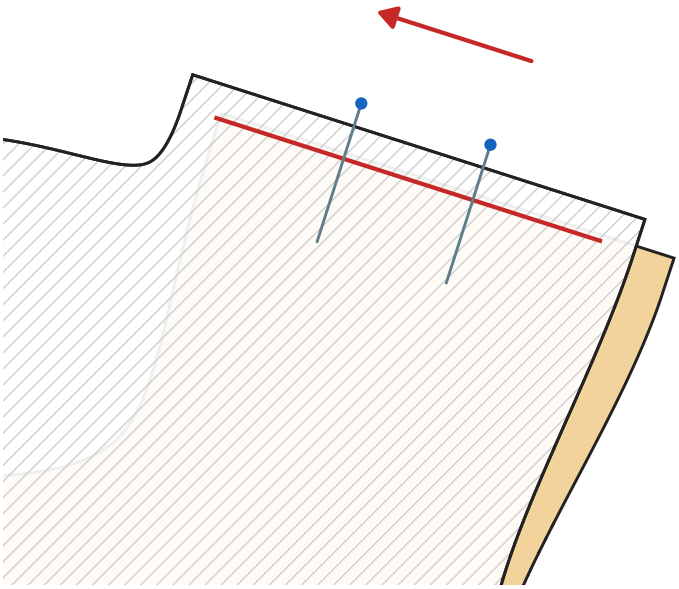
1. Sewing machine: jersey/stretch needle 75–80, polyester thread.
2. Sew a test seam on two layers of a scrap.

Tip: See the basics page at the front of the booklet.



## 5 Sew the shoulder seams

Sewing machine · Pins



1. Lay front and back right sides together, shoulder edges level, and pin.
2. Sew with 10 mm seam allowance from the neckline to the armhole.
3. Press the allowances towards the back.

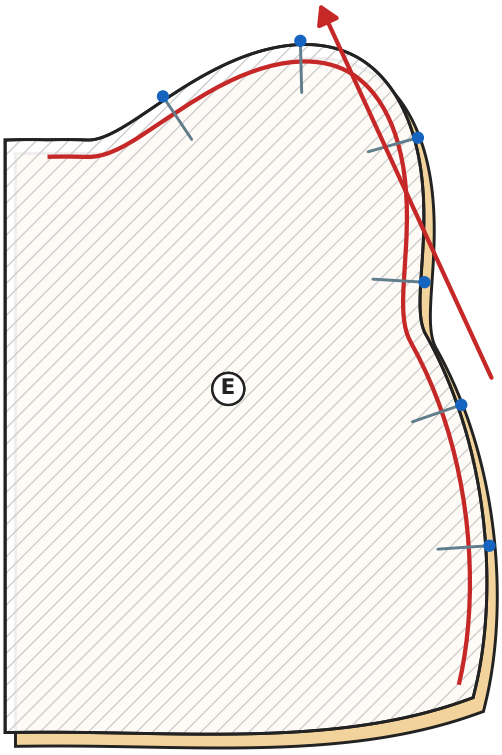
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: Knit stretches at the shoulder: sew a strip of clear elastic or twill tape into the seam.

2×

## 6 Sew the hood centre seam

Sewing machine · Pins



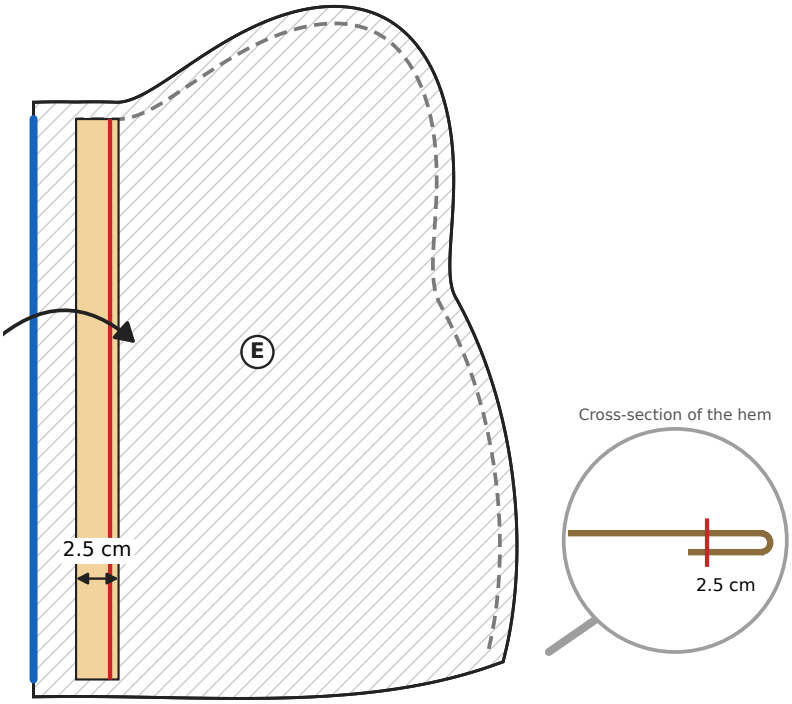

1. Lay the two hood pieces right sides together, curved back edges level, and pin from the neck edge upwards.
2. Sew with 10 mm seam allowance.
3. Press the allowances to one side.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: Topstitch 6 mm beside the seam through the allowances: the seam stays flat.

## 7 Hem the hood front edge

Sewing machine



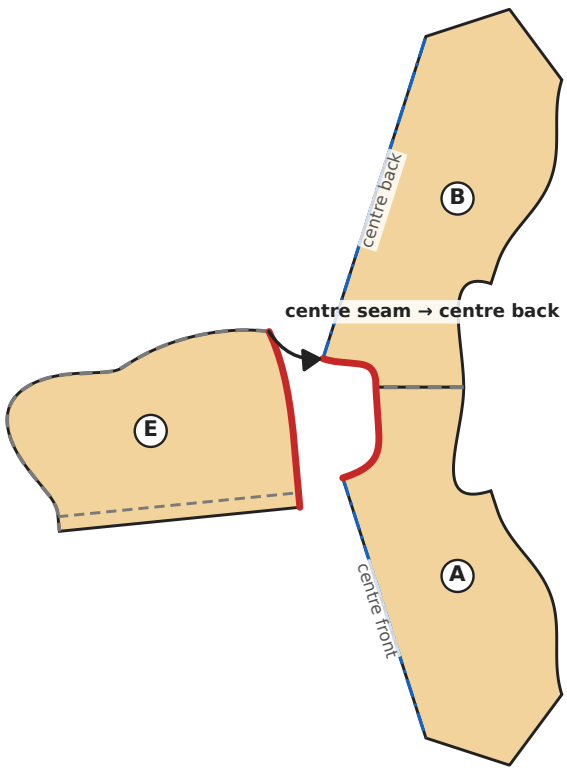

1. Fold the front edge 2.5 cm to the wrong side along the fold line and press.
2. Topstitch 2 cm from the fold, close to the raw edge
3. Leave the ends open for a drawstring.

Sewing machine: twin needle 4 mm, stitch 3 mm

Alternative: Lined hood (option hood_lining): no hem, the lining finishes the front edge and the hood is warmer.

## 8 Sew the hood on

Sewing machine · Pins

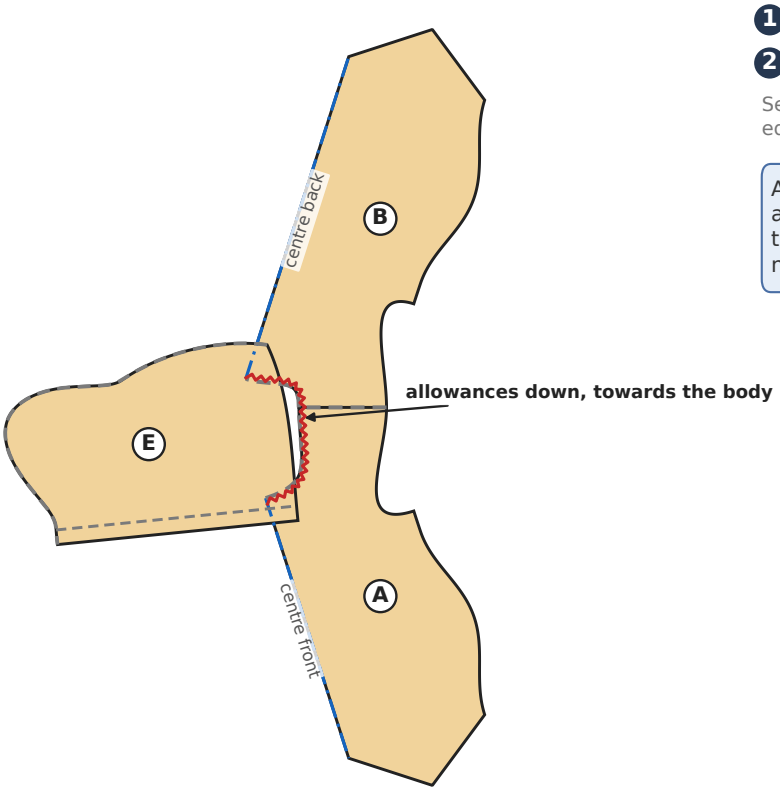



1. Body opened out flat, right side up.
2. Lay the hood right side down onto the neckline, neck edges level.
3. Hood centre seam on centre back, hood notches on the shoulder seams
4. Front edges meet at centre front (overlap 1 cm).
5. Sew all round with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

## 9 Finish the neck seam

Sewing machine



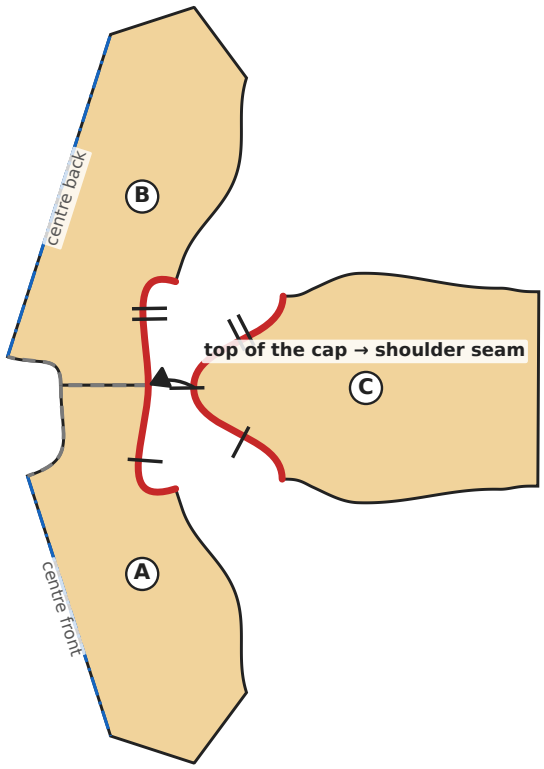

1. Finish both allowances together.
2. Press them down towards the body.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge

Alternative: Neatest finish: cover the allowance with a strip of jersey or cotton tape from shoulder to shoulder over the back neck and stitch it down on both edges.

## 10 Set in the sleeves

Sewing machine · Pins




1. Open the body out flat, right side up.
2. Lay the sleeve right side down on the armhole: top of the cap on the shoulder seam, single notch on the front, double notch on the back.
3. Pin from the middle to both ends, easing in the extra length of the cap.
4. Sew with 10 mm seam allowance, sleeve underneath
5. Press the allowances towards the sleeve.

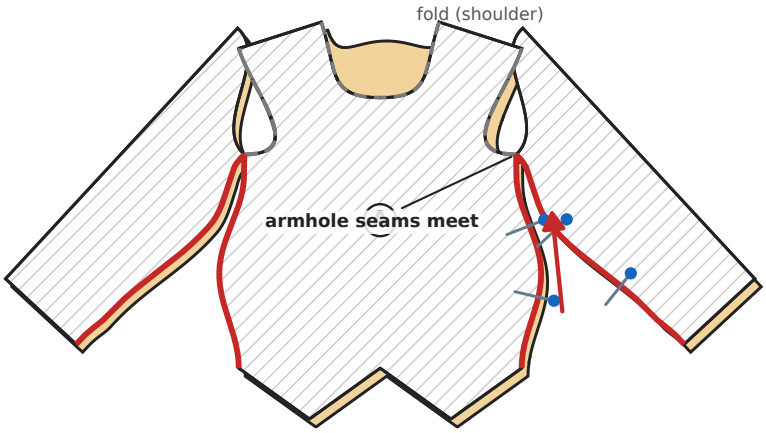
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Sleeve on the bottom: the feed dogs ease the cap in without pleats.

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## 11 Sew side and sleeve seams

Sewing machine · Pins




1. Fold the garment right sides together at the shoulders, the sleeves folded in half lengthwise.
2. Pin side and sleeve seam so that the armhole seams meet exactly at the underarm.
3. Sew in one go with 10 mm seam allowance, from the hem over the underarm to the sleeve hem.
4. Press the allowances towards the back.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Alternative: Cuffs first: sew the cuffs on while the sleeve is still flat, then sew side, sleeve and cuff seam in one go. Easier on small sizes.

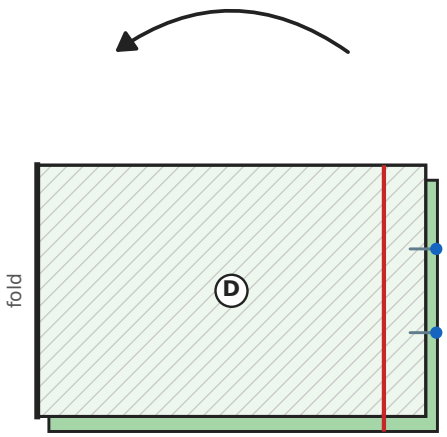
2×

## 12 Close the cuffs into rings

Sewing machine · Pins

1. Fold each cuff crosswise right sides together, short edges level.
2. Sew with 10 mm seam allowance
3. Finger-press the allowances open.

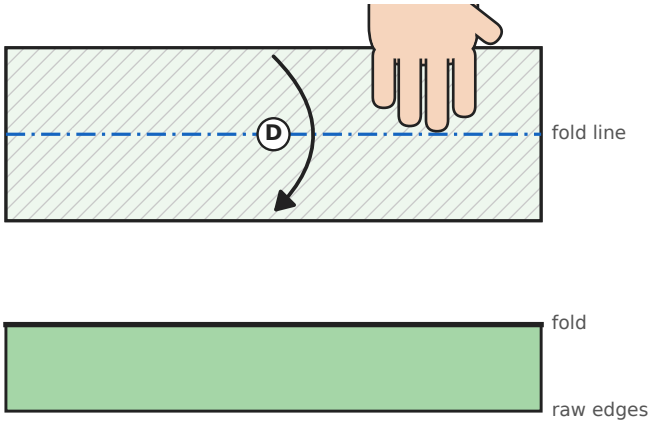
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)




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## 13 Fold the cuffs in half

Iron




1. Fold the ring lengthwise wrong sides together, raw edges level.
2. The seam allowance is now inside
3. The right side shows all round.

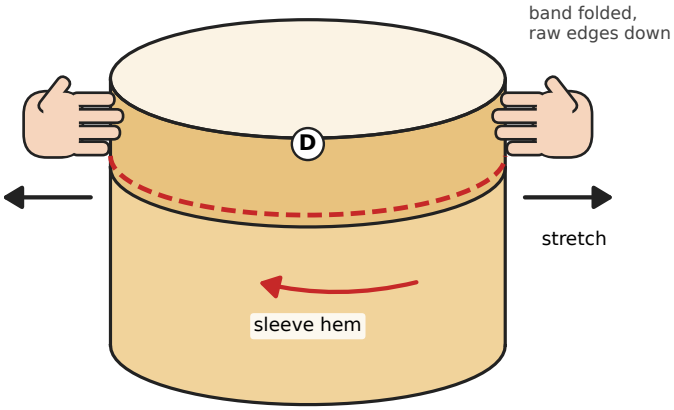
Iron: steam

2×

## 14 Sew the cuffs on

Sewing machine

Pins




1. Turn the sleeve right side out.
2. Slide the cuff over the sleeve end, raw edges level, cuff seam on the sleeve seam
3. Pin there and at the opposite folds of cuff and sleeve.
4. Sew with the cuff on top, stretching the cuff (not the sleeve) to the length of the sleeve between two pins.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: One hand in front of the presser foot, one behind it: both stretch evenly.

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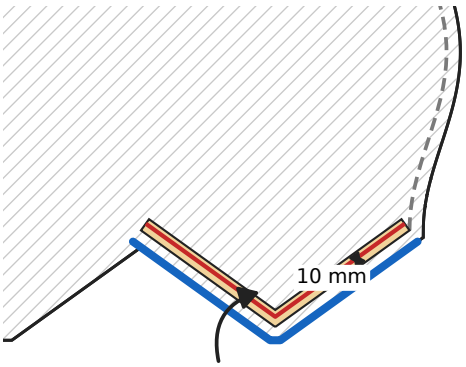
## 15 Sew the hem

Sewing machine

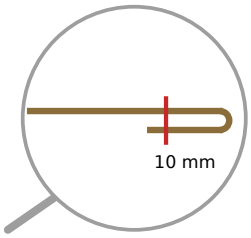
1. Fold the hem 10 mm to the wrong side and press.
2. Sew it down 6 mm from the fold, from the right side.

Sewing machine: twin needle 4 mm, stitch 3 mm

⚠ Important: Shaped hem: clip into every inner point up to 2 mm before the fold and press each point separately.

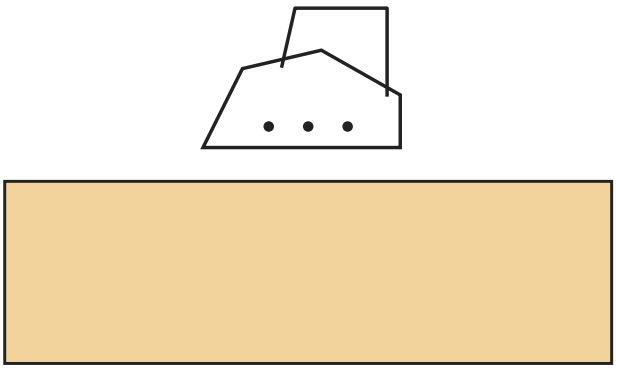



Cross-section of the hem




## 16 Final pressing

1. Press all seams with steam
2. Do not press down hard on sweat.



# Done!



Front view



Back view

After the test garment: Try the test garment on the child: length, chest width, sleeve length and the neckline. Adjust the measurements or options and print the pattern again before you cut the good fabric.

# Starfish hoodie · size 128

wide fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

## Files

- PDF A4 tiles to glue (15 pages + overview): `starfish_hoodie_size128_a4_tiled.pdf`
- Preview A4 tiles: `starfish_hoodie_size128_a4_preview.png`
- SVG master 1:1: `starfish_hoodie_size128_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

## Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

## Body measurements

| Body measurement | Value | Source |
|---|---|---|
| Body height | 128.0 cm | table |
| Chest | 64.0 cm | table |
| Waist | 57.0 cm | table |
| Hip | 69.0 cm | table |
| Head | 52.5 cm | table |
| Neck | 29.0 cm | table |
| Inseam | 57.5 cm | table |
| Side length | 76.0 cm | table |
| Arm length | 50.0 cm | table |
| Back length | 31.0 cm | table |
| Shoulder width | 10.4 cm | table |
| Thigh | 38.6 cm | table |
| Ankle | 18.6 cm | table |

## Finished measurements (virtually sewn)

| Finished measurement | Value |
|---|---|
| Chest width | 75.0 cm |
| Hem width | 95.8 cm |
| Length from neck point | 47.6 cm |
| Sleeve length | 50.0 cm |
| Sleeve opening | 18.6 cm |
| Upper-arm width | 25.0 cm |
| Neckline circumference | 48.9 cm |
| Hood height | 33.0 cm |

## Pattern pieces

| Piece | Cutting | Fabric | Seam allowance |
|---|---|---|---|
| Front | cut 1 on fold | Main fabric | seam allowance 1.0 cm included |
| Back | cut 1 on fold | Main fabric | seam allowance 1.0 cm included |

| Piece | Cutting | Fabric | Seam allowance |
|---|---|---|---|
| Sleeve | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included |
| Sleeve cuff | cut 2 | Rib knit | seam allowance 1.0 cm included |
| Hood | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included; Front edge (casing/facing): 2.5 cm |

## Fabric needed

- **Main fabric:** about 1.05 m at 150 cm width (single layer, incl. 10 % for shrinkage; folded would need 1.25 m)
- **Rib knit:** about 0.15 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Design: option hem_band = False (instead of True): needed for the shape.
- Design: option kangaroo = False (instead of True): needed for the shape.
- Creative design 'Starfish': the shape deviates from the base block on purpose; see 'Design' in the validation report.
- Shaped hem: finish it with a narrow hem (overlock or zigzag the edge, turn 1 cm, topstitch) or a facing; clip into every inner point.

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

**Cut on fold** (Front, Back): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

## Validation report

### Sewability check

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Side seam front ↔ back | 30.3 cm | 30.3 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Armhole ↔ sleeve cap (ease) | ease 0–8 mm | 36.5 cm ↔ 36.8 cm | +3.2 mm | ✓ OK | Sweat / French Terry: 0–8 mm |
| Seam pairs | Shoulder seam front ↔ back | 11.9 cm | 11.9 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Sleeve seam (front ↔ back) | 35.6 cm | 35.6 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Sleeve cuff ↔ sleeve hem (stretched) | 70–85 % of 26.5 cm | 18.6 cm = 70 % | -79.5 mm | ✓ OK | rib band is stretched while sewing |
| Seam pairs | Hood ↔ neckline (each side) | 24.5 cm | 24.5 cm | +0.0 mm | ✓ OK | |
| Seam pairs | Hood centre seam (left ↔ right) | 57.2 cm | 57.2 cm | +0.0 mm | ✓ OK | |
| Angles/transitions | Underarm: side seam ↔ armhole front | 90° | 90.4° | +0.4° | ✓ OK | |
| Angles/transitions | Underarm: side seam ↔ armhole back | 90° | 90.5° | +0.5° | ✓ OK | |
| Angles/transitions | Shoulder ↔ armhole (front + back) | 180° | 179.5° | -0.5° | ✓ OK | |
| Angles/transitions | Sleeve cap top (smooth) | 180° | 178.2° | -1.8° | ✓ OK | |
| Angles/transitions | Sleeve underarm ↔ cap front | 90° | 90.9° | +0.9° | ✓ OK | |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Angles/transitions | Shoulder ↔ neckline (front + back) | 180° | 180.3° | +0.3° | ✓ OK | |
| Angles/transitions | Neckline at centre front | 90° | 90.2° | +0.2° | ✓ OK | |
| Angles/transitions | Neckline at centre back | 90° | 89.6° | -0.4° | ✓ OK | |
| Angles/transitions | Hem at centre front | 126° | 125.7° | +0.0° | ✓ OK | changed by the design (intended) |
| Angles/transitions | Hem ↔ side seam front | 126° | 126.1° | -0.0° | ✓ OK | changed by the design (intended) |
| Angles/transitions | Hood: front edge ↔ neck seam | 90° | 90.0° | +0.0° | ✓ OK | |
| Measurements | Chest circumference pattern = body + ease | 74.9 cm | 75.0 cm | +0.1 % | ✓ OK | body 64.0 cm + +17 % ease |
| Measurements | Neckline (fabric edge) fits over the head | ≥ 52.5 cm | 61.2 cm | +86.7 mm | ✓ OK | neckline incl. stretch ≥ head circumference |
| Measurements | Hand fits through the sleeve cuff | ≥ 19.8 cm | 33.4 cm | +135.7 mm | ✓ OK | hand circumference ≈ 1.55 × wrist |
| Measurements | Sleeve length plausible | 45.0 cm – 62.5 cm | 50.0 cm | +0.0 mm | ✓ OK | arm length 50.0 cm |
| Measurements | Body length plausible | 38.4 cm – 79.4 cm | 47.6 cm | +0.0 mm | ✓ OK | from the neck point |
| Measurements | Hood tall enough | ≥ 28.9 cm | 33.0 cm | +40.8 mm | ✓ OK | hood height ≥ 0.55 × head circumference |
| Geometry | Front: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Front: grainline & cutting | present | cut 1 on fold | | ✓ OK | Main fabric |
| Geometry | Back: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Back: grainline & cutting | present | cut 1 on fold | | ✓ OK | Main fabric |
| Geometry | Sleeve: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Sleeve: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Sleeve cuff: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included |
| Labels | Sleeve cuff: grainline & cutting | present | cut 2 | | ✓ OK | Rib knit |
| Geometry | Hood: outline & seam allowance | closed, SA present | ok | | ✓ OK | seam allowance 1.0 cm included; Front edge (casing/facing): 2.5 cm |
| Labels | Hood: grainline & cutting | present | cut 2 (mirrored pair) | | ✓ OK | Main fabric |
| Geometry | Hood: turned allowance lies flat | ≤ 0.5 mm | 0.0 mm | | ✓ OK | Front edge (casing/facing): corners mirrored at the fold line |
| Geometry | Front: cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.2° off 90° | | ✓ OK | Centre (fold) |
| Geometry | Back: cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.4° off 90° | | ✓ OK | Centre (fold) |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Labels | Front: notches | on the cutting line | 1 notch mark(s) | | ✓ OK | |
| Labels | Back: notches | on the cutting line | 2 notch mark(s) | | ✓ OK | |
| Labels | Sleeve: notches | on the cutting line | 4 notch mark(s) | | ✓ OK | 1 at corners |
| Design (intended) | Side seam: bulge +3.0 cm at 55 % | base block | +0.0 … +3.0 cm | | ✓ OK | intended deviation (design 'Starfish') |
| Design (intended) | Sleeve seam: profile +3.0 cm at 30 %, +0.5 cm at 85 % | base block | +0.0 … +3.0 cm | | ✓ OK | intended deviation (design 'Starfish') |
| Design (intended) | Hood centre seam: peak +9.0 cm at 70 % | base block | +0.0 … +9.0 cm | | ✓ OK | intended deviation (design 'Starfish') |
| Design (intended) | Hem: 2 × points, 7.0 cm deep | base block | +0.0 … +7.0 cm | | ✓ OK | intended deviation (design 'Starfish') |
| Freedom of movement | Side seam: girth after the design | ≥ 60.8 cm | 75.0 cm | +0.1 cm vs. base | ✓ OK | not tighter than the base block (at 8 %) |
| Freedom of movement | Sleeve seam: girth after the design | ≥ 17.9 cm | 26.3 cm | +1.4 cm vs. base | ✓ OK | not tighter than the base block (at 8 %) |
| Freedom of movement | Hem: girth after the design | ≥ 60.0 cm | 95.8 cm | +18.0 cm vs. base | ✓ OK | not tighter than the base block (at 100 %) |
| Design sewability | Side seam: steepness of the shape | ≤ 1.0 | 0.36 | | ✓ OK | offset change per length along the seam (1.0 = 45°) |
| Design sewability | Sleeve seam: steepness of the shape | ≤ 1.0 | 0.50 | | ✓ OK | offset change per length along the seam (1.0 = 45°) |
| Design sewability | Hem: points can be sewn and turned | ≥ 60° | 114° | | ✓ OK | outer points of the reshaped edge |
| Design sewability | Hem: inner corners | ≥ 60° | none | | ✓ OK | |

**Result:** 51 OK, 0 warning(s), 0 error(s): the pattern is sewable.

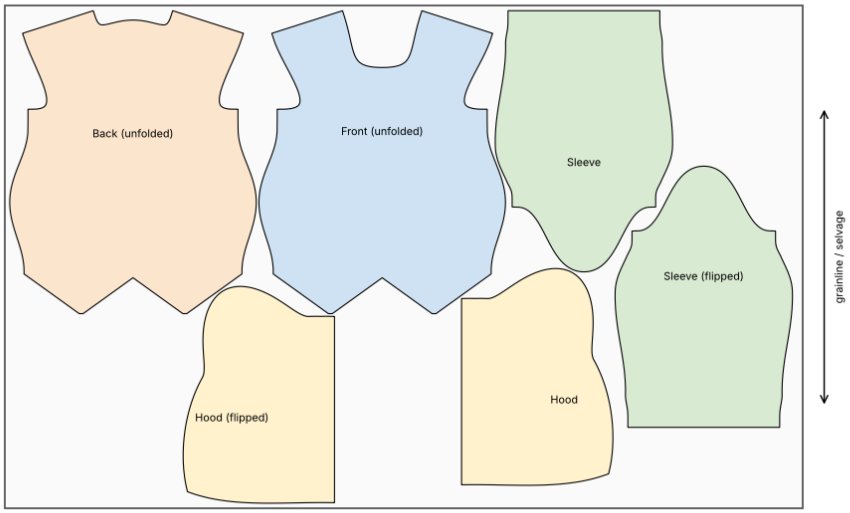
## Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"chest_width": 749}}`).

| Name | Value | Derived from | Note |
|---|---|---|---|
| `chest_width` | 74.9 cm | chest × (1 + 17 % ease) | |
| `neck_width_half` | 8.1 cm | neck / 5 + 1.5 cm | |
| `front_neck_depth` | 10.4 cm | neck_width_half + 1.5 cm | |
| `back_neck_depth` | 1.5 cm | fixed | |
| `shoulder_length` | 11.9 cm | shoulder + allowance for fit 'wide' | |
| `shoulder_slope` | 17 | fixed | |
| `armhole_depth` | 18.2 cm | 18 % of chest_width + 2.5 cm + 3 % | |
| `body_length` | 47.6 cm | back length + 13 % of height | |

Cutting layout · Starfish hoodie size 128

Main fabric: 150 cm wide, single layer
needed ≈ 1.05 m




Rib knit: 70 cm wide, single layer
needed ≈ 0.15 m

